HEAT TRANSFER ENGINEERING
2025, VOL. 46, NOS. 9–10, 852–864
https://doi.org/10.1080/01457632.2024.2347178

Taylor & Francis
Taylor & Francis Group

Check for updates

Effect of Using Fractals as Grain Configuration on the Performance of Solid Rocket Motors

Soham Bhattacharya[a] and Raghav Mundhra[b]

[a]Mechanical Engineering Department, Jadavpur University, Kolkata, India; [b]Mechanical Engineering Department, Indian Institute of Technology, Kharagpur, India

ABSTRACT

Grain design is a very crucial consideration for determining the performance of solid propellant rocket motors. A rocket's performance and combustion characteristics (thrust profile, chamber pressure variation, propellant mass flow rate, and sliver among other important performance parameters) depend very strongly on the grain configuration. We propose fractal shapes (Koch star, Koch snowflake, Gosper, and Cesaro) as new port geometries and theoretically predict their performance using simulations based on the fast-marching method implemented in the open-source software Open-Motor. The initial thrust for fractal geometry grains was found to be significantly higher than the commonly used industry standard BATES and Finosyl ports. The fractal grains also showed a significant decrease in sliver as compared to the Finocyl port that is generally used to increase performance as compared to the circular BATES ports. Especially, The Koch snowflake port had a drastically lower sliver. These new grain geometries proposed in this work can be used for applications in which low initiation and ignition times along with high initial thrust are crucial. Moreover, these geometries can improve the performance of the solid rocket motor (SRM) owing to their better regression characteristics which promise complete utilization of the entire grain web.

Introduction

Solid rocket propulsion is an established technology and has been used in many space missions. Due to their simple design and operation, compactness, ability to operate quickly, long shelf life, and no leakage concerns, solid rocket motors (SRMs) are often the preferred choice for military applications [1].

Unlike liquid propulsion systems, the solid propellant (called grain) is already stored within the combustion chamber of solid propellant rocket motors. There is a central cavity called the port, through which hot pressurized combustion gases flow out toward the nozzle to generate the thrust. The propellant burns on the exposed surfaces of this internal cavity to generate hot combustion gases. The combustion proceeds along the normal to the burn surface. Hence, the initial shape of the port is the variable that directly influences the regression shape of the grain, this in turn determines the amount of propellant burnt at any moment which is critical to the creation of chamber pressure, generation of thrust and impulse and every other performance parameter of the motor. Thus, the geometric configuration of the port has an immense impact on the performance of the motor [1, 2].

The grain geometry is designed based on the flight mission requirements. Owing to the developments in solid propellant rocket technology, various grain configurations, as shown in Figure 1, along with their performance characteristics are available to motor designers, for example, the space shuttle's redesigned SRM used an 11-pointed star grain [2].

Currently, BATES (circular port, cylindrical grain), Finocyl (circular port with slots or fins), and star are commonly used as port shapes. BATES grains are simply hollow cylinder/tubular grains, i.e., the central cavity is circular and extremely simple and easy to manufacture. However, the initial internal area of the propellant that is exposed is extremely small and this low regression rate leads to slower growth of chamber pressure leading to lower initial thrust and higher time for initiation. To solve this problem, often, radial slots of needed widths (in Finocyl type) are introduced, or the port shape is changed to star to increase the burning area. While this results in some limited

CONTACT Raghav Mundhra rag.mundhra@gmail.com Mechanical Engineering Department, Indian Institute of Technology, Kharagpur 721302, India
© 2024 Taylor & Francis Group, LLC

Nomenclature

a	Temperature coefficient
A	Surface Area, m^2
Al	Aluminum Powder
AP	Ammonium Perchlorate
ATK	Alliant Tech Systems Inc.
FMM	Fast marching method
HTPB	Hydroxyl-terminated polybutadiene
I_{sp}	Delivered specific impulse, s
Kn	Ratio of propellant burn area to the nozzle throat area
$\dot{m}$	Mass flow rate, kg/s
n	Burning rate pressure exponent
p	Pressure, MPa
r	Burning rate, mm/s
SRM	Solid rocket motor
T	Thrust, N
v	Velocity, m/s

Greek Symbols

ρ	Density, kg/m^3

Subscripts

atm	Atmospheric
b	Burn Area
cb	Chamber
e	Exit
m	Momentum
p	Pressure

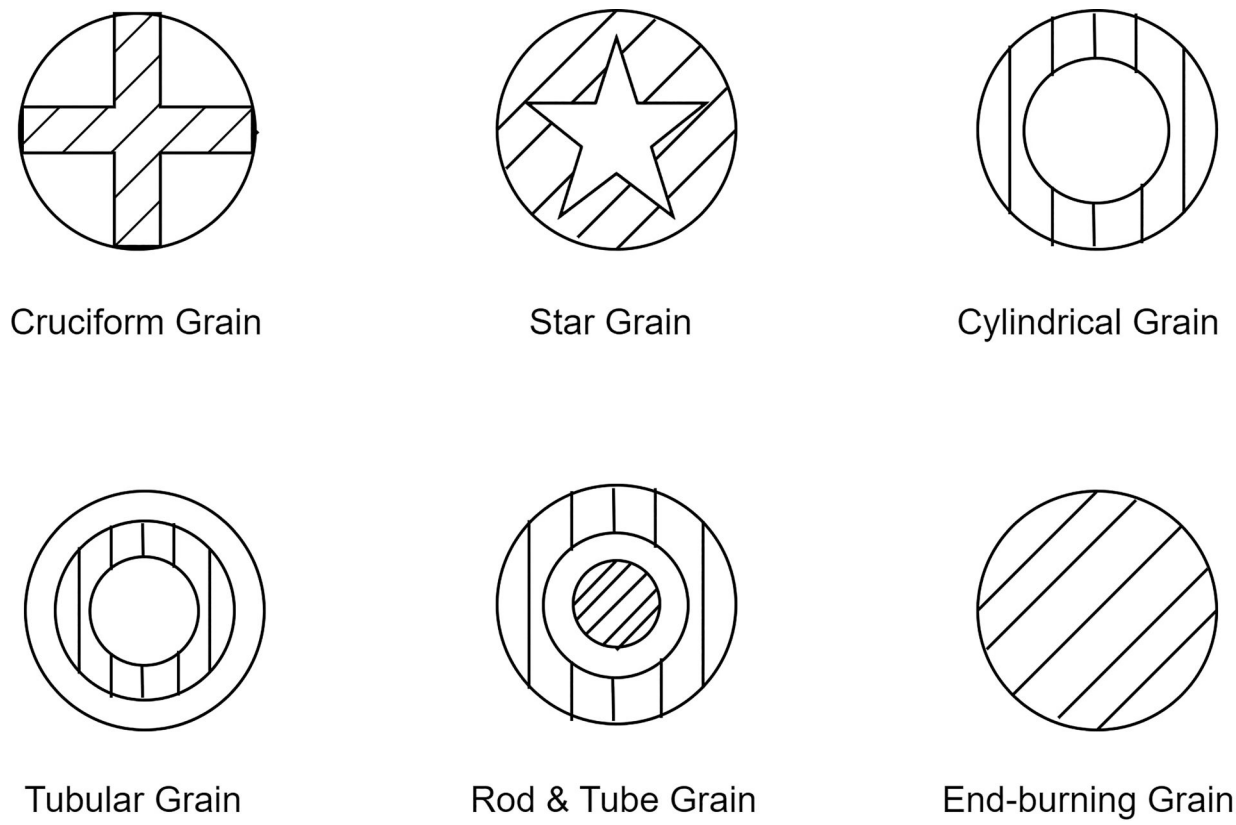

Figure 1. Cross section of different grain configurations used in rocket motors, adapted from Sforza [2].

increase in thrust and pressure, especially at the initial stages, it severely increases residual, unburnt propellant (called sliver) remains in the motor periphery which reduces the performance efficacy of the motor [1].

Several analytical and numerical methods have been developed for the design and analysis of different grain geometries [3–7]. Many optimization studies have been carried out in order to choose the optimum grain geometry based on the performance requirement of the motor [8–10]. Nisar *et al.* [11] carried out the optimization of the geometric variables of a 3D Finocyl grain to satisfy the requirements of propellant mass, thrust, and burning time. Recently, Tianfang *et al.* [12] optimized a single 'wagon wheel' grain for a hybrid rocket motor. In the last few decades, a lot of work has been done on the design, analysis, and optimization of grain geometry leading to its evolution from simple cylindrical to complex geometries.

Heat transfer characteristics in the nozzle are dependent on the propellant ballistics [13]. The propellant ballistics, in turn, also depends on the grain design. Ahmad [14] performed a 3D axisymmetric simulation of the initial grain of the re-usable SRM to theoretically predict the convective heat transfer rate in the nozzle. The nozzle heat transfer characteristics need to be critically analyzed during the design of the SRM. Grain design plays a crucial role in the heat transfer rates in the nozzle.

Advancement in manufacturing technologies, especially 3D printing, has made it possible to make objects with complicated geometry. Funami *et al.* [15] showed it is possible to manufacture and use such complicated geometries in rocket motor grains. They analyzed the effects of using a star-shaped complicated port and compared the local regression rates and thrust characteristics with that of a circular (BATES) port. It was observed that the fractal port generated more thrust and mass flow rate due to better regression when compared to the circular (BATES) port.

Fractals are highly complex self-repeating geometries made out of copies of themselves such that they appear similar at every scale. It may be noted that the word 'fractals' has evolved from its classic definition of self-similar shapes. Now, fractality also implies irregularity and fractal dimension of a curve or contour and is interpreted as a measure of its roughness. This has paved the way for fractals to be used for description of complex dynamic time series by multifractal detrended analysis [16]. Fractals now find applications in description analysis and prediction of various natural phenomena in various fields ranging from ecology [17] to combustion [18, 19].

Through the careful design of the grain geometry, the performance of rocket motors can be controlled and modulated as per the mission requirements. Although a catalog of numerous grain configurations is already available to the rocket propulsion engineer, still work is going on to explore more geometries so that optimum performance of the motor can be achieved in alignment with the mission requirements. In this work, we propose new port geometry in which the grain configuration is based on different 2D fractal shapes. In order to analyze the impact of these highly complex geometries on SRM performance, simulations of the motor with these port geometries have been performed using an open-source software. The performance of rocket motors having simple port geometry (BATES and Finocyl) is compared with that of rocket motors having fractals as their port geometry through these simulations. Theoretical prediction of the performance of SRMs with such grain design has not been done in any previous work. Importance is given to residual propellant mass (sliver ratio), thrust, and pressure profiles. The parameters like propellant (taken as AP/HTPB/Al – Ammonium perchlorate/Hydroxyl-terminated polybutadiene/Aluminum powder – which has been used to validate the simulator), propellant mass, outer grain diameter and length, ambient conditions, nozzle designs, and some other relevant parameters have been kept constant while only altering the port shape.

Numerical strategy

The thrust resulting from grain regression characteristics for different grain configurations has been evaluated and compared using the fast-marching method [20, 21] in the open-source internal ballistic simulator openMotor. Unlike other SRM simulators, openMotor allows the use of custom grain configurations which is imperative for the study as the grains being compared are novel and do not conform to existing ones. Other than thrust, rocket motor performance parameters like chamber pressure, ratio of propellant burn area to the nozzle throat area (Kn), propellant mass, propellant mass flow rate, and residual mass were also calculated and compared using this software. It should be noted that openMotor considers instant ignition of the entire internal exposed port surface and thus ignores the small ignition delay that usually exists in ignition of actual rocket motors. Further, only simple axisymmetric conical nozzles can be created for simulations, support for commonly used real-world nozzles like the bell nozzle is not present in the package.

The internal ballistic analysis of an SRM is done using computational fluid dynamics methods involving moving boundaries. As the propellant undergoes combustion, the control volume continually expands. This expansion occurs in all three dimensions and local changes in all the parameters under observation must be taken into consideration for accurate results. Moving boundaries may be handled by a Lagrangian (interface tracking) or an Eulerian (interface capturing) approach. In the Lagrangian approach the nodes and discretized elements follow the moving surface while in the Eulerian approach the nodes and discretized elements are stationary. Fast marching method (FMM) is based on the level-set method which follows an Eulerian or interface capturing method. FMM is a specific case of the level-set method where the moving wall has a positive sign only, as in an SRM. In FMM, the interface capturing is done according to the local

speed variations following which the domain is sliced with the help of this interface and the boundary conditions provided during pre-processing by the user are applied at these surfaces created by the slicing. This is followed by coupling and iterative solving of various governing equations and flow conditions as specified during pre-processing. FMM is preferable to other methods as it handles the complicated geometric interfaces in fractals very accurately.

Figure 2. Grain configuration used in openMotor for validation of ATK Star™ 27 motor.

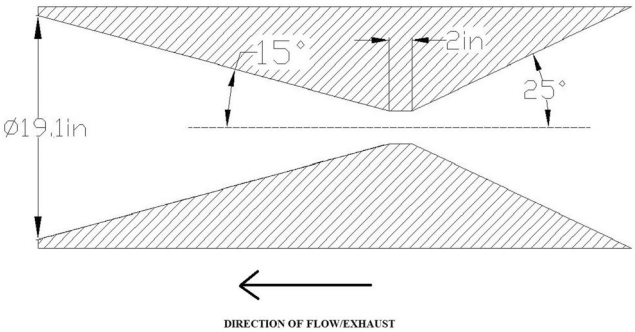

Figure 3. Nozzle configuration used in openMotor for validation of ATK Star™ 27 motor.

Validation of openMotor

The accuracy of the FMM algorithm in openMotor was validated by simulating the performance of the Alliant Tech Systems Inc. (ATK) Star™ 27 motor which was developed in 1975 for the Canadian Communications Research Centre's Communications Technology Satellite. The grain, nozzle, and propellant design along with ambient conditions were replicated in openMotor and the outputs of simulation were compared with the performance data available. The outer diameter of the motor was 0.69 m and the overall length of the motor was 1.23 m. The grain was configured as an eight-pointed star with a web of 0.21 m or 8.17 inches, web fraction was 6% volumetric loaded was 92.6% and the average burning area was 0.89 m^2. The grain design is shown in Figure 2 and the nozzle design is shown in Figure 3. The performance parameters and combustion characteristics are shown in Table 1 and Figure 4, respectively, were recorded by simulating vacuum conditions during operation.

AP/HTPB/Al in a ratio of 72:12:18 was used as the propellant. The thermodynamic properties for openMotor were derived from the CEARUN calculator and the methods used in Stephenson [22] and Sutton [1] as density: 1775 kg/m^3; burning rate exponent: 0.28; burning rate co-efficient: 0.0404 in/s-psin;

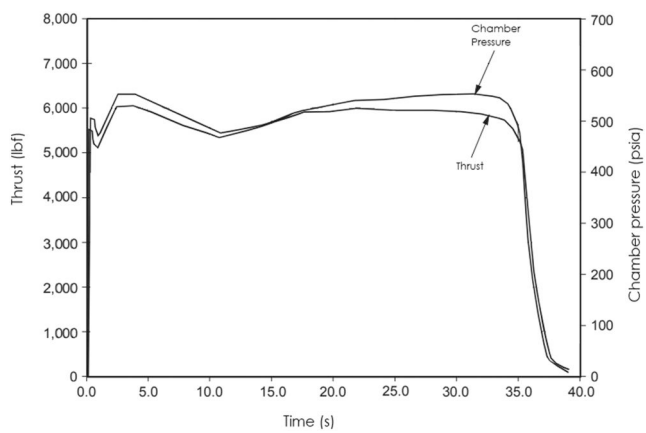

Figure 4. Variation of thrust and chamber pressure with time in actual ATK Star™ 27 motor.

Table 1. Comparison of combustion characteristics and parameters in actual ATK Star™ 27 motor with outputs from openMotor simulated motor.

Parameter	Actual motor	Simulated motor	Percentage difference
Burn time/action time, s	34.4	35.5	+3.2%
Propellant weight, kg	333.6	331.2	−0.73%
Action time, s	37.3	39.09	+4.8%
Burn time averaged Kn	233.56	234.38	0.35%
Action time average chamber pressure, MPa	3.461	3.447	−0.4%
Maximum chamber pressure, MPa	3.923	3.978	+1.40%
Total impulse, kN·s	951.3	916.2	−3.69%
Delivered specific impulse, N·s/kg	2820.65	2795.5	−0.83%

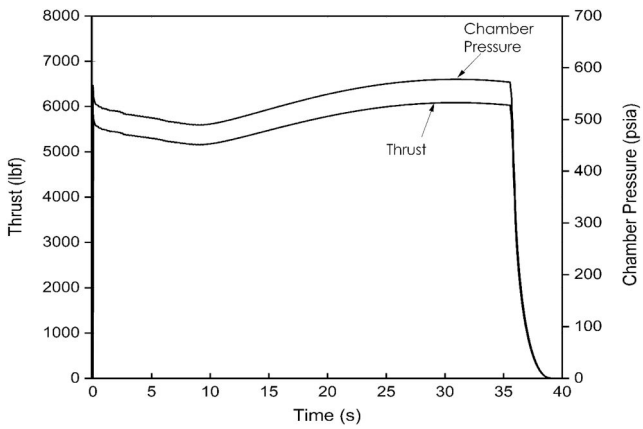

Figure 5. Variation of thrust and chamber pressure with time in motor simulated in OpenMotor.

flame temperature: 3265 °C; molar mass: 28.9 g/mol; characteristic velocity: 1560 m/s. Total mass of the propellant was 333.7 kg. Initial throat diameter was equal to 0.07 m; initial exit diameter was 0.48 m resulting in an expansion ratio of 48.6:1. The throat area was 0.0038 m^2. Kn 233.56 on average. Nozzle exit cone half angle was 15°.

As is seen from Figures 4 and 5 thrust and pressure variation for the actual motor and the one simulated in openMotor are almost equivalent barring a small deviation in the very early stage which is attributed to the small amount of time the actual motor took to reach steady state and the time taken by the entire inner port surface to ignite. Table 1 further establishes the accuracy of the software as parameters only vary by minute amounts.

Fractals as port geometry

The performance of rocket motors with Koch star and Koch snowflake ports were compared with the two commonly used industrial port geometries: BATES, which is a simple circular port and Finosyl, which is similar to the BATES port but with radial fins or slots. The dimensions and general configuration of the BATES grain is shown in Figure 6(a) and the dimensions and general configuration for the Finocyl port is shown in Figure 6(b).

Total propellant mass and outer diameter were kept constant at 24.72 kg and 0.148 m respectively for all port geometries used irrespective of generic and fractal ports. Similarly, grain length was kept constant at 0.838 m. As mentioned, the BATES port is a circular central cavity and thus results in a tubular grain. The diameter of the cavity was 0.03 m. Only the top surface was inhibited, and the bottom surface of the grain was free to burn, this is similar to an actual motor where the surface near the nozzle is free to burn whereas the other end of the grain is inhibited due to the casing.

The Finocyl port consists of a circular central cavity with radial slots/fins of desired width. The purpose of these slots is to increase the initial burning area and subsequent regression of the grain to improve thrust, pressure, and regression rates during burning. The circular cavity on which the port is based is 1.18, there are 4 radial slots each of length 0.009652 m and width 0.003048 m. Same as the BATES grain, only the bottom surface was free to burn whereas the top surface of the grain was inhibited.

Construction of Koch curve-based fractals for port geometry

Any fractal based on the Koch curves is made by transforming each straight-line segment into 4-line segments. Each of these smaller line segments is equal to each other and is one-third of the length of the original segment.

The 5-pointed Koch star fractal port

The Koch snowflake fractal was constructed by transforming each of the sides of a star made by five points to Koch curves. This resulted in an increase in the burning area from BATES grain while the port area remained similar. The grain with its dimensions, in inches, is shown in Figure 6(c).

The Koch snowflake fractal port

The Koch snowflake fractal was constructed by transforming each side of an equilateral triangle to Koch curves. This resulted in an increase in the burning area from BATES grain while the port area remained similar. The grain with its dimensions, in inches, is shown in Figure 6(d).

The Cesaro fractal port

The Cesaro fractal was constructed by transforming each of the 8 sides of a four-pointed star to Koch curves. The new lines were again transformed into another set of Koch curves. This resulted in an increase in the burning area from BATES grain while the port area remained similar. The grain with its dimensions, in inches, is shown in Figure 6(e).

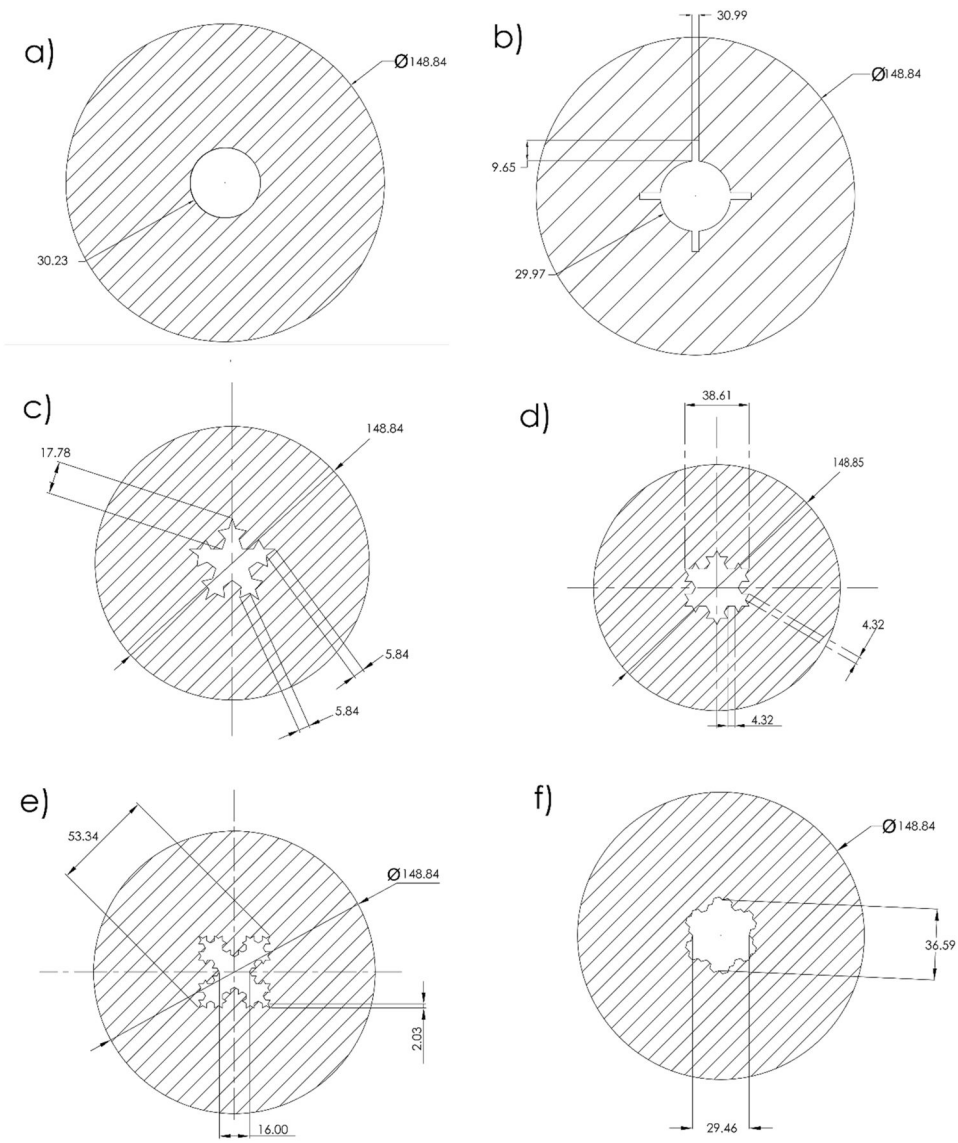

Figure 6. Various port geometry used for comparison in the study (dimensions are in inches).

The Gosper fractal port

The construction of the Gosper fractal is different from the above fractals as it isn't created by the Koch transformation rule. This fractal is made by dividing each line into three lines that are not all equal in length. The Gosper fractal port grain with its dimensions, in inches, is shown in Figure 6(f).

Propellant description

Simulated fuel was AP/HTPB/Al composite fuel, with characteristics similar to that used in ATK Star^{TM} 27 motor.

Nozzle description

Initial throat diameter was taken to be 0.043 m, exit diameter was kept at 0.112 m resulting in an expansion ratio of 6.83. Convergence and divergence cone half angles were taken as 25° and 15°, respectively. Figure 7 shows the geometry of the nozzle used for the simulation.

Results and discussion

In this section, we discuss the differences observed in thrust, chamber pressure, Kn, and sliver mass (residual propellant mass) when using the different geometries described above as grain ports. As mentioned, all other parameters like propellant mass, outer grain diameter, and nozzle design were kept constant for the different ports.

The mass of gases produced by combustion at any point of time, $\dot{m}$, is given in terms of propellant burn area at that point, A_b; burning rate of

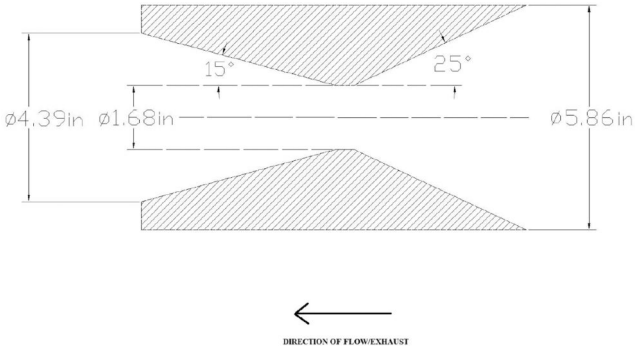

Figure 7. Geometry of the nozzle used for the simulation.

the propellant, r and the density of the propellant, ρ by:

$$\dot{m} = A_b \times r \times \rho \tag{1}$$

For a given propellant ρ is constant while r only depends on chamber pressure. As a result, especially in the initial ignition condition, only the port geometry decides $\dot{m}$ and higher burn area results in higher gas production rates which reduce delay in initiation.

Figures 8–11 offer a visual representation of the progressive regression of a cross-section of the grains at key moments in time along with the mass of propellant left at these times. The inner surface of the port along with the bottom surface of the grain undergoes combustion to generate hot gases which then generate pressure and thrust. The fractal geometry has the ability to drastically increase burning surface without reducing the propellant mass that can be loaded as seen by Figures 12(a,b). Figure 12(a) shows the mass flow rate out of the motor with respect to time. At any moment, the values of ejected mass are significantly higher for fractal ports. Figure 12(b) shows that while the initial propellant mass loaded into the motor is the same for all cases, the deterioration rate of the propellant is higher for fractal ports. Hence, the fractal ports offer significantly higher mass flow and regression rates which are a result of better burn surface offered by such novel ports. Figure 12(c) offers an insight into Kn, the ratio between grain burn area and nozzle exit area at various points of time. The fractal ports have much better burn area to offer which can be seen by their higher Kn ratios. This results in an increase in the amount of propellant that can be burned which results in quick pressure build up in the chamber which results in an increased mass flow rate, $\dot{m}$, offering a great increase especially in the initial phase of initiation which helps build up the chamber pressure faster and reduces initiation time. This increase in $\dot{m}$ also improves thrust greatly as discussed next.

The total thrust, T generated by a SRM is made of two components: momentum thrust, T_m and pressure thrust, T_p. The momentum thrust is due to exit of pressurized hot gases from the chamber and simple control volume analysis gives its value as the product of the propellant mass flow rate, $\dot{m}$, and its exhaust velocity relative to the vehicle, v. The pressure thrust on the other hand is due to imbalances between the existing atmospheric pressure, p_{atm} and the nozzle exit pressure, p_e and is given by the product of the differences in the pressure and cross-sectional area of the nozzle at exit, A_e. Thus, we have:

$$T = T_m + T_p \tag{2}$$

$$T_m = \dot{m}v \tag{3}$$

$$T_p = (p_e - p_{atm})A_e \tag{4}$$

Combining,

$$T = \dot{m}v + (p_e - p_{atm})A_e \tag{5}$$

Fractal ports with the improved $\dot{m}$ owing to their ability to increase burn area which offers a direct increase to the amount of matter that can be ejected out of the nozzle, thereby increasing momentum thrust. Due to faster build up, the pressure rise in the chamber is also much faster and higher, this can be seen in Figure 12(d). As p_e increases, for the same p_{atm}, the pressure thrust increases. The combined increase of both thrust components improves thrust characteristics of SRMs fitted with these novel ports seen in Figure 12(e).

As seen in Figure 12(d) the fractal ports offer faster build-up of pressure as a result of which the p_{cb} increase is much better than generic ports. The burn rate of a propellant, r, is related to p_{cb} as:

$$r = a \times (p_{cb})^n \tag{6}$$

Here, a is the temperature coefficient what is shown, (an empirical constant) and n is the burning rate exponent. The value of n is less than 1.0 for stable combustion. As a grain port develops a higher p_{cb}, the burning rate increases which in turn causes faster regression and higher rate of consumption of the propellant, this further increases the developed pressure and thus the burning rate is further increased. Fractal ports benefit greatly from this positive feedback effect. Obviously, this increase also results in higher mass flow rates and thus, higher momentum thrust. As the growth of p_{cb} in fractal ports is much better than generic ports, the feedback effect happens much faster in them, and this helps in exponentially better regression of the grain as compared to generic ports.

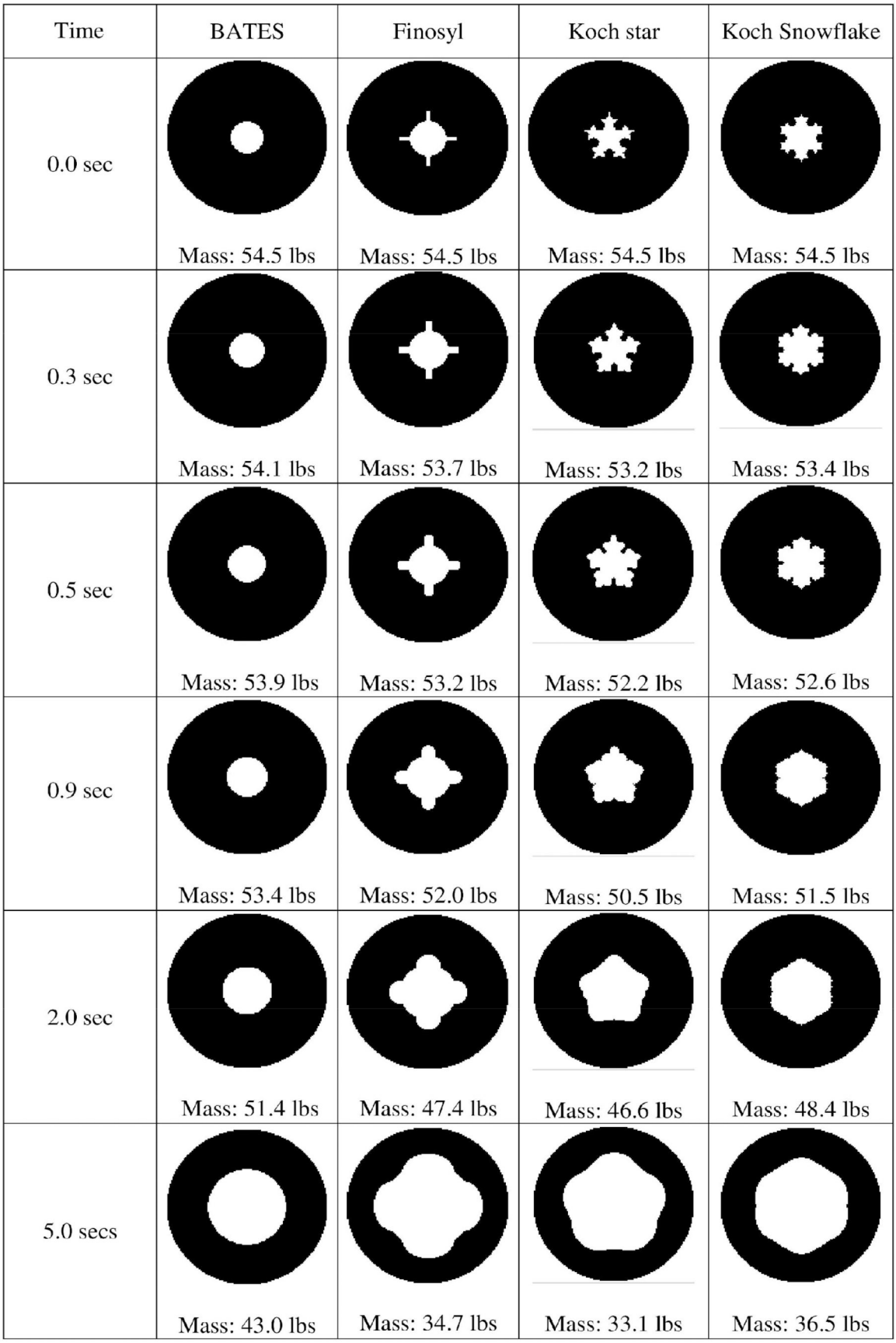

Figure 8. Comparison of regression of grain between generic, Kock star, and Koch snowflake port shapes for 0 to 5 s.

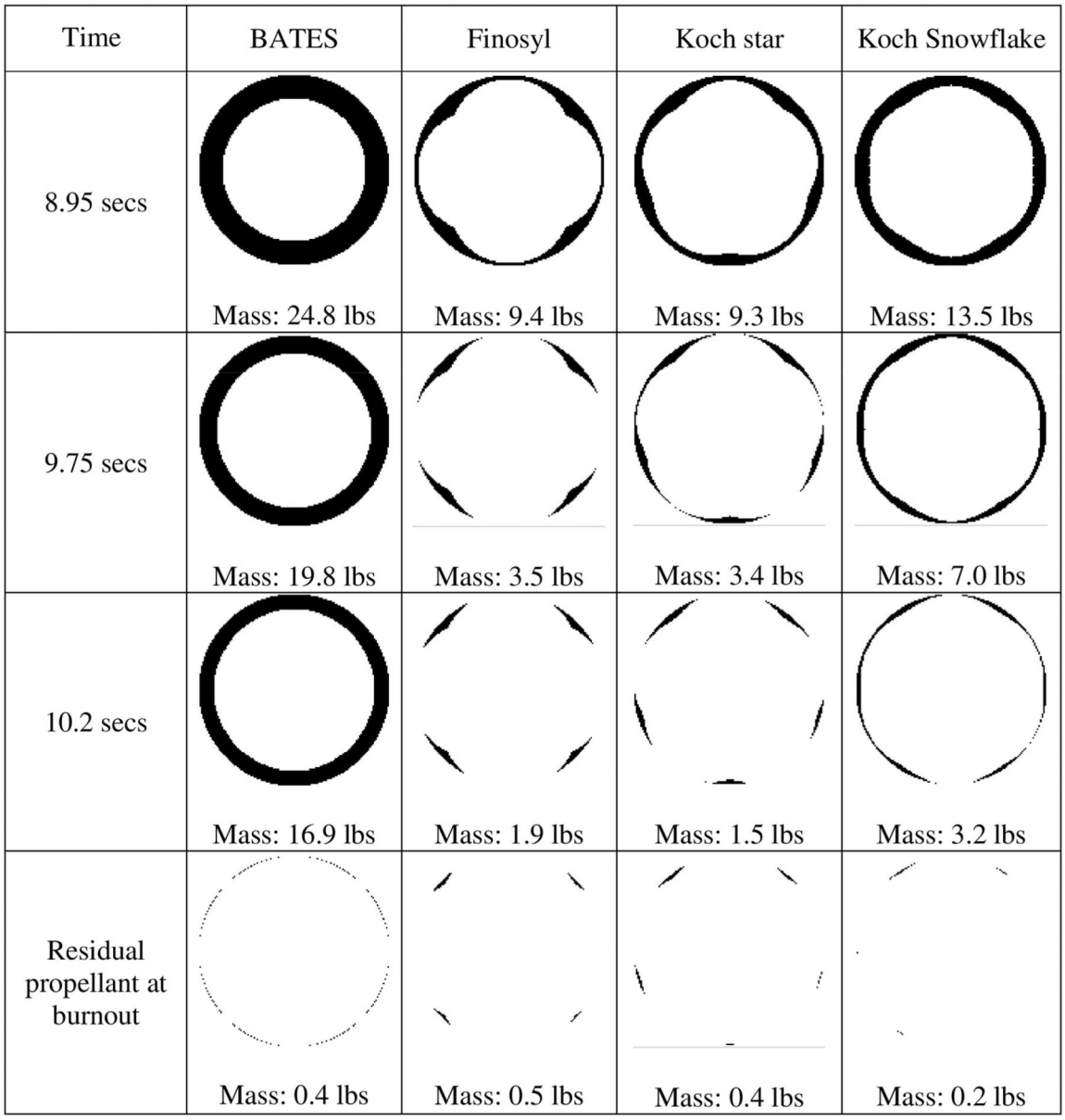

Figure 9. Comparison of regression of grain between generic, Kock star, and Koch snowflake port shapes for 8.95 s to burnout.

The thrust against time curves shown in Figure 12(e) shows using fractal ports drastically improve initial thrust generated. For comparison, the initial time averaged thrust for BATES and Finocyl ports are 810.51 and 2570.32 N respectively while for Gosper fractal, Koch snowflake, Koch star, and Cesaro fractal ports it rises to 1774.35, 3080.88, 4350.80, and 4400.22 N respectively. This is a direct consequence of higher regression rates shown in Figure 12(f) and further establishes the superiority of using fractal ports in grains. For comparison, the initial time averaged regression rates for BATES and Finocyl ports are 0.0036 and 0.00439 m/s, respectively while for Gosper fractal, Koch snowflake, Koch star, and Cesaro fractal ports it rises to 0.0041, 0.0046, 0.0049, and 0.0049 m/s, respectively.

While the Finosyl port offers limited improvement over BATES port in terms of thrust and regression, it results in a marked increase in unburnt propellant residue, or sliver fraction by 35.8% increase as shown in Figure 13. This results in a decrease in motor efficiency and is highly undesirable. Fractal ports, on the other hand, offer an elegant solution by keeping the increase down to a minimal 4.9%, in the case of Koch star port and even reducing the residue by a staggering 41.4% in the case of Koch snowflake port. The Gosper and Cesaro fractal ports also showed a reduction by 15.4% and 23.1%, respectively.

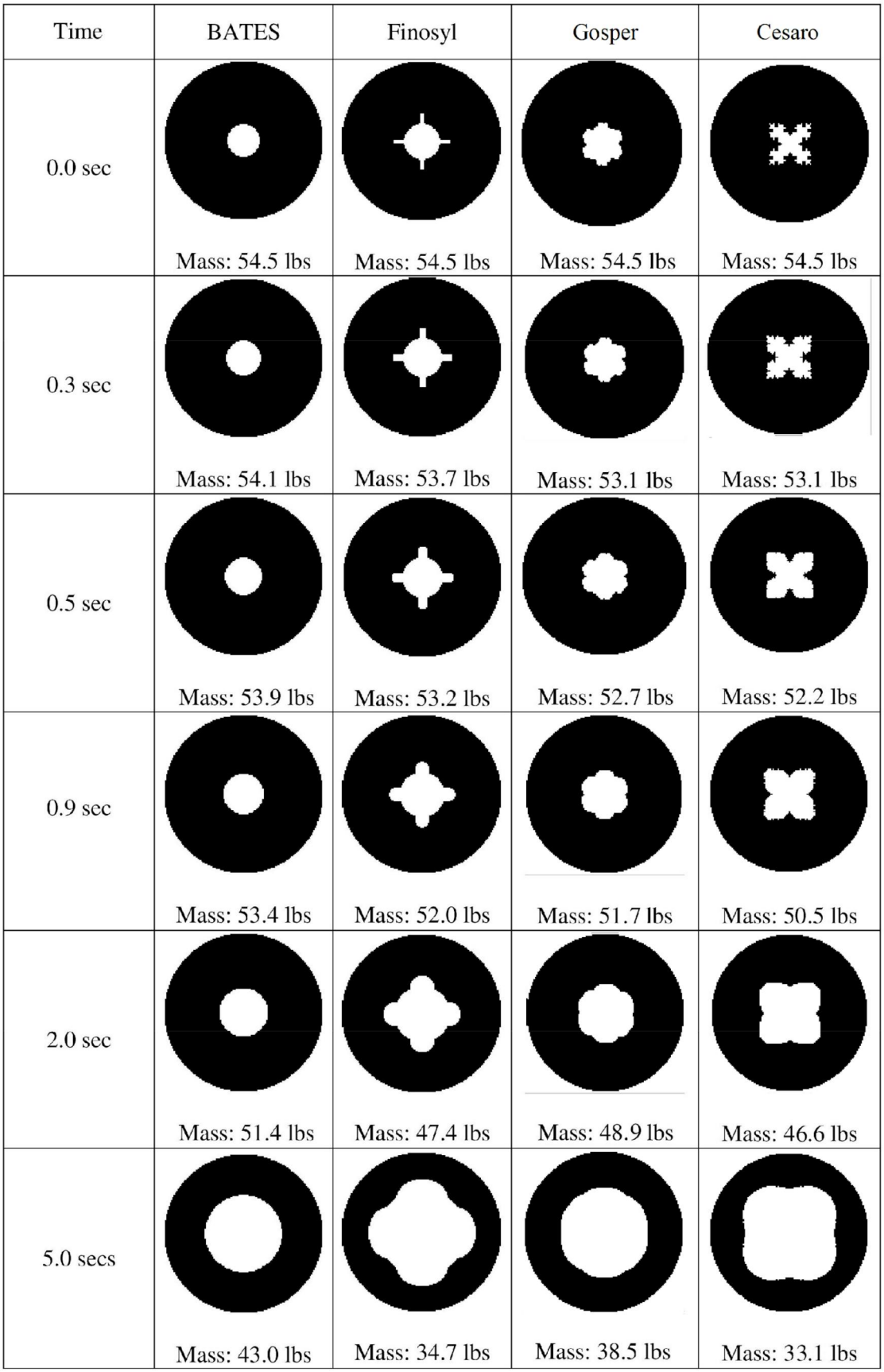

Figure 10. Comparison of regression of grain between generic, Cesaro, and Gosper port shapes for 0 to 5 s.

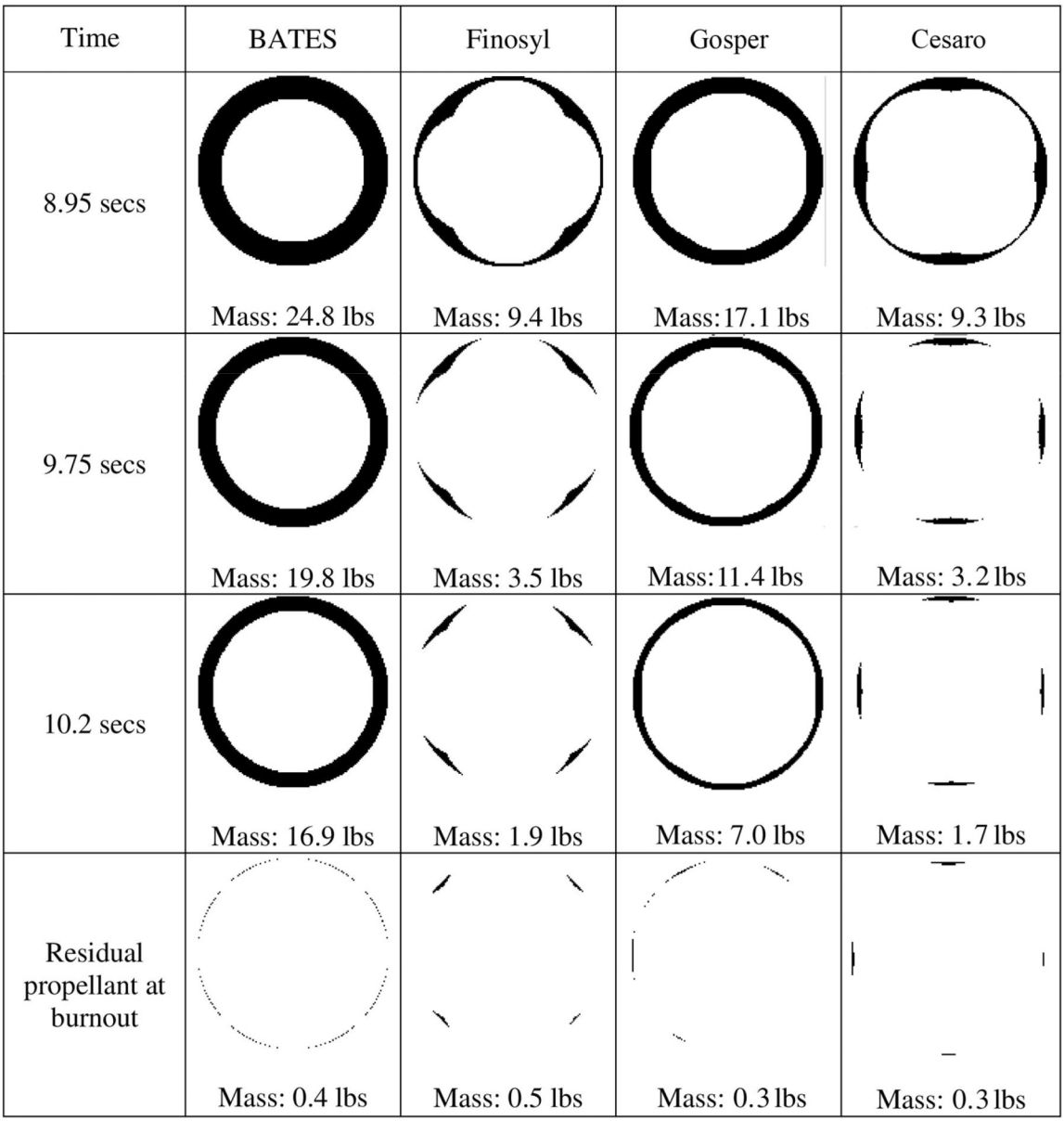

Figure 11. Comparison of regression of grain between generic, Cesaro, and Gosper port shapes for 8.95 s to burnout.

The total thrust generated by the grains is compared using the delivered specific impulse, Isp, metric, i.e., total impulse generated divided by propellant mass. Finosyl grain had the worst delivered Isp at 216.78 s. The Koch star fractal port had a 1.5% better delivered Isp at 219.96 s. The BATES grain had a delivered Isp of 217.87 s, around 1% lower than the Koch star fractal. The Koch snowflake, Cesaro, and Gosper fractal ports had a delivered Isp of 218.69, 218.64, and 217.55 s, respectively. The generally better delivered Isp of the fractal ports can be attributed to better sliver, resulting in more complete utilization of fuel.

Conclusions

Fractal shapes have been proposed to be used as grain configuration for SRMs. The performance of a solid propellant rocket motor with fractal grain port geometries was compared with that of prevalent grain port geometries. The motors with fractal ports performed significantly better than the motors with circular and finned ports. The initial thrust in Gosper fractal, Koch snowflake, Koch star, and Cesaro fractal ports were 2.18, 3.80, 5.36, and 5.42 times that of the commonly used BATES port. The chamber pressure and mass flow rates increased faster in fractal ports and thus can be

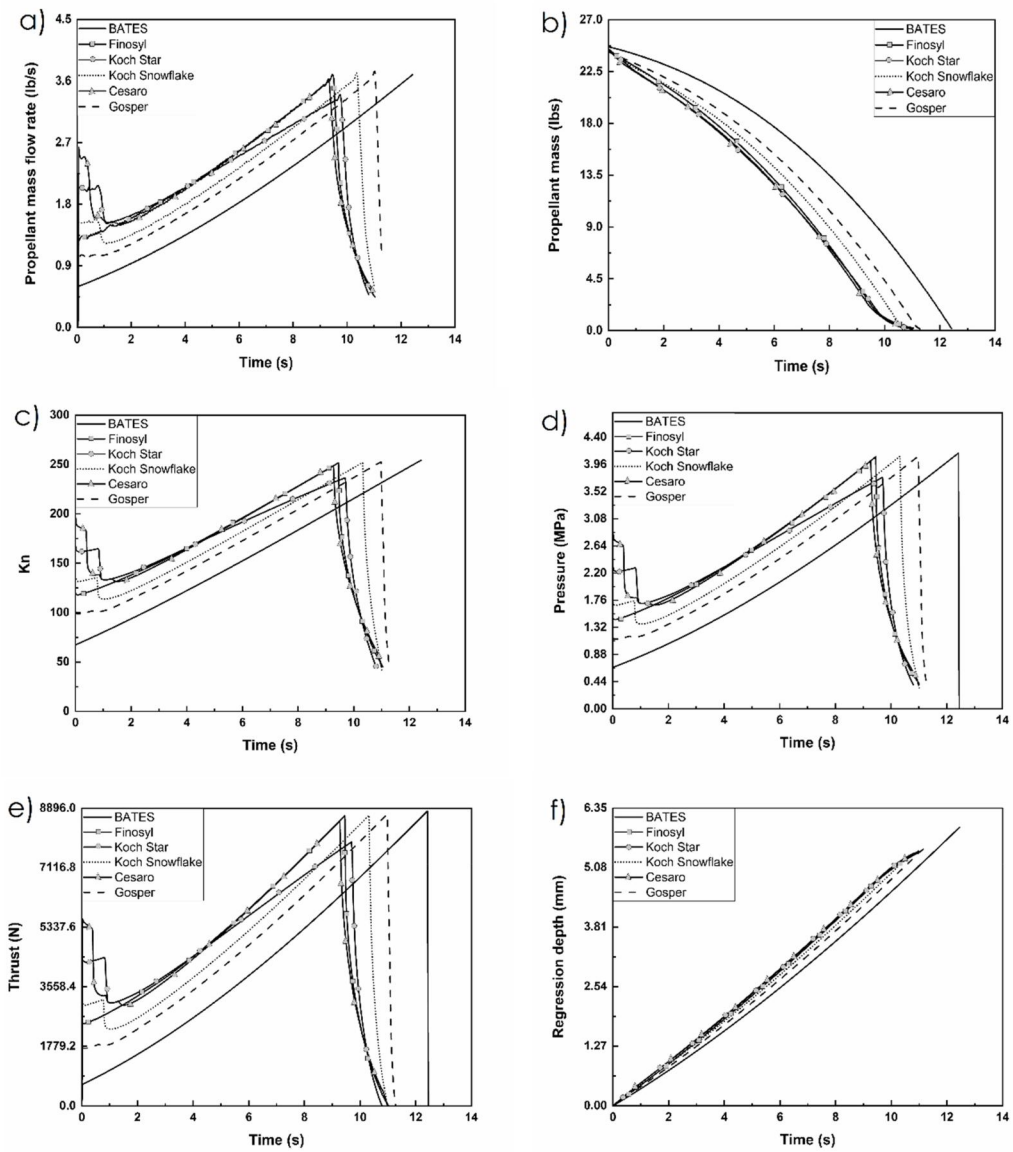

Figure 12. Comparison of combustion characteristics mass with time between different port shapes.

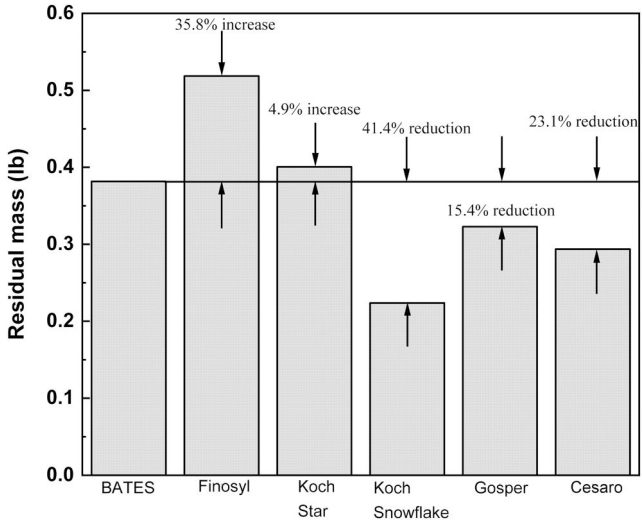

Figure 13. Comparison of sliver or residual mass between different port shapes.

used for faster ignition and thrust generation in real world applications. Regression of the grain also increased by 1.26 times in the Cesaro fractal port, 1.13 times in the Gosper fractal port, 1.35 times in Koch star port, and 1.26 times in Koch snowflake port when compared to the BATES port. Further, fractal ports do not suffer from the drawback of high residual propellants like the Finosyl port. In fact, the increase observed in Koch star port was a mere 4.9% compared to the 35.8% increase observed in Finocyl port while the Gosper and Cesaro fractal showed a decrease in residual mass of 15.4% and 23.1%, respectively from the BATES port. The Koch snowflake drastically improved residual mass, offering a decrease of 41.4% residue from the BATES port. The faster regression capabilities of fractal ports along with the higher mass flow rates observed can drastically improve the drawbacks of hybrid

rockets, which suffer and perform inadequately due to low regression and mass flow rates.

Disclosure statement

No potential conflict of interest was reported by the authors.

Notes on contributors

Soham Bhattacharya is a third-year undergraduate student at the Department of Mechanical Engineering, Jadavpur University, Kolkata, India. His primary research interests lie in the field of aerospace propulsion, combustion stability, and fluid mechanics. He is a part of a research project studying the multifractal analysis of combustion stability.

Raghav Mundhra finished his B.E. in Mechanical Engineering from Jadavpur University, Kolkata, India in 2019. Currently, he is pursuing his MS in Mechanical Engineering from Indian Institute of Technology, Kharagpur, India. His primary research interests lie in the fields of aerospace propulsion, fluid mechanics, and thermodynamics. He has authored three conference papers.

References

[1] G. P. Sutton and O. Biblarz, *Rocket Propulsion Elements*, 9th ed. Hoboken, NJ, Wiley, 2017.

[2] P. M. Sforza, *Theory of Aerospace Propulsion*, 2nd ed. Cambridge, MA, Elsevier, 2017.

[3] R. J. Hejl and S. D. Heister, "Solid rocket motor grain burnback analysis using adaptive grids," *J. Propuls. Power*, vol. 11, no. 5, pp. 1006–1011, Sep–Oct. 1995. DOI: 10.2514/3.23930.

[4] F. Mashayek, H. Farzad and N. Ashgriz, "A geometry independent technique for solid propellant grain design," *Proc. Inst. Mech. Eng. G J. Aerosp.*, vol. 210, no. 3, pp. 209–220, Jul. 1996. DOI: 10.1243/PIME_PROC_1996_210_365_02.

[5] R. Hartfield, R. Jenkins, J. Burkhalter and W. Foster, "A review of analytical methods for solid rocket motor grain analysis," presented at the 39th AIAA/ASME/SAE/ASEE Jt. Propuls. Conf. Exhib., Huntsville, AL, Jul. 20–23, 2003. DOI: 10.2514/6.2003-4506.

[6] M. A. Willcox, M. Q. Brewster, K. C. Tang and D. S. Stewart, "Solid propellant grain design and burnback simulation using a minimum distance function," *J. Propuls. Power*, vol. 23, no. 2, pp. 465–475, Mar–Apr. 2007. DOI: 10.2514/1.22937.

[7] C. Tola and M. Nikbay, "Internal ballistic modeling of a solid rocket motor by analytical burnback analysis," *J. Spacecr. Rocket*, vol. 56, no. 2, pp. 498–516, Jan. 2019. DOI: 10.2514/1.A34065.

[8] S. Acik, "Internal ballistic design optimization of a solid rocket motor," M.S thesis, Dept. Mech. Eng., Middle East Tech. Univ., Ankara, Turkey, 2010.

[9] M. Johannsson, "Optimization of solid rocket grain geometries," M.S thesis, Dept. Aerosp. Eng., KTH R. Inst. Technol., Stockholm, Sweden, 2012.

[10] C. Tola and M. Nikbay, "Multidisciplinary optimization of a solid propellant sectional geometry for internal ballistic and structural strength criteria," presented at the 52nd AIAA/SAE/ASEE Jt. Propuls. Conf., Salt Lake City, UT, Jul. 25–27,2016. DOI: 10.2514/6.2016-5001.

[11] K. Nisar, L. Guozhu and Q. Zeeshan, "A hybrid optimization approach for SRM Finocyl grain design," *Chin. J. Aeronaut.*, vol. 21, no. 6, pp. 481–487, Dec. 2008. DOI: 10.1016/S1000-9361(08)60164-8.

[12] W. Tianfang *et al.*, "Optimization design in single wagon-wheel fuel grain of hybrid rocket motor," *FirePhysChem*, vol. 1, no. 4, pp. 212–221, Dec. 2021. DOI: 10.1016/j.fpc.2021.11.010.

[13] R. A. Ahmad, "Discharge coefficients and heat transfer for axisymmetric supersonic nozzles," *Heat Transf. Eng.*, vol. 22, no. 6, pp. 40–61, 2001. DOI: 10.1080/014576301317048424.

[14] R. A. Ahmad, "Convective heat transfer in the reusable solid rocket motor of the space transportation system," *Heat Transf. Eng.*, vol. 26, no. 10, pp. 30–45, 2005. DOI: 10.1080/01457630500248554.

[15] Y. Funami, K. Uchishima, S. Homme, S. Nishino and A. Takano, "Evaluation of hybrid rocket fuel grain with a star fractal port using combustion experiments," *Aerosp. Technol. Jpn.*, vol. 19, no. 3, pp. 295–303, 2021. DOI: 10.2322/tastj.19.295.

[16] J. W. Kantelhardt *et al.*, "Multifractal detrended fluctuation analysis of nonstatinary time series," *Phys. A Stat. Mech. Appl.*, vol. 316, no. 1–4, pp. 87–114, Dec. 2002. DOI: 10.1016/S0378-4371(02)01383-3070.

[17] G. Sugihara and R. M. May, "Applications of fractals in ecology," *Trends Ecol. Evol.*, vol. 5, no. 3, pp. 79–86, Mar. 1990. DOI: 10.1016/0169-5347(90)90235-6.

[18] V. R. Unni and R. I. Sujith, "Multifractal characteristics of combustor dynamics close to lean blowout," *J. Fluid Mech.*, vol. 784, pp. 30–50, Dec. 2015. DOI: 10.1017/jfm.2015.567.

[19] S. Chatterjee, M. Das, A. Mukhopadhyay and S. Sen, "Experimental investigation of breakup of annular liquid sheet in a hybrid atomizer," *J. Propuls. Power*, vol. 38, no. 6, pp. 1–10, Jun. 2015. DOI: 10.2514/1.B35574.

[20] J. A. Sethian, "A fast marching level set method for monotonically advancing fronts," *Proc. Natl. Acad. Sci. USA*, vol. 93, no. 4, pp. 1591–1595, Feb. 1996. DOI: 10.1073/pnas.93.4.1591.

[21] J. A. Sethian, "Fast marching methods," *SIAM Rev.*, vol. 41, no. 2, pp. 199–235, Jun. 1999. DOI: 10.1137/S0036144598347059.

[22] J. Stephenson, "Design of nozzle for high-powered solid rocket propellant," *Undergrad. J. Math. Model. One + Two*, vol. 9, no. 1, pp. 1, Oct. 2018. DOI: 10.5038/2326-3652.9.1.4895.

Copyright of Heat Transfer Engineering is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.